Human Finitude:
AI and the Foundation of Civilization

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Abstract

Politics, science, the economy, and morality all rest on a presupposition that has held throughout human history and was therefore treated as a permanent fact: no one so far surpasses everyone else that the others can neither restrain nor understand them. Human institutions were built by finite beings for finite beings, and they work only as long as the presupposition holds. Now both the effort to build AI that exceeds human beings across most domains and the advance of AI capability are running toward removing the presupposition. We argue that institutions built on rough equality among people would end only in dictatorship if AI capability were concentrated in a few, or in lasting instability if it spread to many. Avoiding both would require institutions that do not presuppose such equality, which have not yet been built and may not be possible.

Introduction

AI is entering original intellectual work: AI systems, often working alongside mathematicians, have produced proofs on open problems in mathematics (OpenAI, 2026b; Alpöge et al., 2026; OpenAI, 2026a). The institutions building AI state their goal openly as systems that exceed human beings across most domains (OpenAI, 2018; Amodei, 2024; Safe Superintelligence Inc., 2024; Altman, 2025), which is also how the field defines its most capable systems (Morris et al., 2024; Bostrom, 2014). Major states have made leadership in AI a national goal (The White House, 2026; State Council of the People's Republic of China, 2017). We argue that the effort now under way, and the trend of capability, run toward removing a presupposition on which human institutions rest.

Concern about catastrophic AI risk has centered on two scenarios, AI acquiring goals of its own and turning against humanity (Bostrom, 2014; Yudkowsky and Soares, 2025; Russell, 2019), and people using AI to cause mass harm (Hendrycks et al., 2023). Both scenarios require that AI first become more capable than human beings. We argue that the step alone, without goals of its own, without loss of control, and without malice, already erodes the foundations of human institutions.

Human institutions presuppose that no one so far surpasses everyone else that the others can neither restrain nor understand them. Put in ordinary words, nobody knows everything and nobody can do everything. Earlier thinkers took the presupposition as a plain fact about human beings and wrote it down without argument (Hobbes, 1651; Hart, 1994; Hume, 1998). The exception they could imagine was a person so far above everyone else that they stand among others like a god among men (Aristotle, 4th century

BCE). They could leave the exception hypothetical because no human being had ever come close to it. The effort to build AI is the first effort aimed at a capability that would make it real.

A few laboratories hold the most capable systems (OpenAI, 2026a), so the exception would arise among people, as a gap between those who command AI and those who do not. How much of their capability reaches everyone else depends on what the holders choose to release. If the capability stayed in the hands of a few, those few would stand beyond the reach of everyone else. If it spread to many, then wherever destruction is easier than protection, each holder could do harm that others could not stop (Bostrom, 2019).

Work on *gradual disempowerment*, part of a wider literature on harms that require neither rogue AI nor deliberate misuse (Zwetsloot and Dafoe, 2019; Christiano, 2019; Kasirzadeh, 2025; Davidson et al., 2025; Drago and Laine, 2025; Bullock et al., 2025), asks whether societies will still need human beings as AI takes over their work (Kulveit et al., 2025). Whether people can still restrain and understand those who command AI is a different condition from whether they are still needed. The two conditions vary independently: a ruler can still need the labor of many people while standing so far above them that they cannot restrain the ruler, and AI can replace human labor while those who control it remain within reach of one another. Escaping the restraint and understanding of everyone else requires a far larger gap in capability than replacing their labor does.

Politics, science, the economy, and morality rest on the same presupposition, so that its failure would reach all four at once. Following the consequences through each domain, the paper finds no non-dictatorial landing point: within institutions built on rough equality among people, the only outcomes left would be lasting instability and dictatorship. Avoiding both would require institutions that do not presuppose such equality, which have not yet been built and may not be possible.

Politics

In every political order, the presupposition takes the form of the weaker side's ability to restrain the stronger. Even rulers holding the heavier weapons have had to make concessions to their subjects, because their rule depended on the cooperation of others (Tilly, 1990; Mosca, 1939; Arendt, 1970).

Two conditions let the weaker side restrain the stronger (Schelling, 2020). The first is that it can reliably make the stronger side pay a cost, as when many people band together against one. The second is that it holds something the stronger side needs and cannot simply seize, such as labor, obedience, or taxes paid without a fight. When at least one condition holds, we call the two sides *comparable*. When neither holds, the weaker side can no longer restrain the stronger.

AI can do the work of the officials and soldiers whose cooperation the strong once needed (Davidson et al., 2025), bringing closer the point at which force no longer needs human hands and the second condition fails (Arendt, 1970). A state whose revenue no longer comes from its citizens likewise has less reason to answer to them (Drago and Laine, 2025; Kulveit et al., 2025). The first condition fails once the gap in capability grows

so large that punishment can no longer reach the stronger party, leaving the weak only the option of attaching themselves to it for protection (Hart, 1994).

In ancient Greek political thought, laws were held to apply only among people roughly equal in birth and ability. Democratic cities used ostracism, a vote that exiled a citizen for a time, against anyone who grew too powerful through wealth or influence. Only a person outstanding in virtue was thought to deserve rule instead of exile (Aristotle, 4th century BCE). Superiority gained through AI is superiority in capability, of the same kind as superiority through wealth or influence, so by the ancient standard a person who towers over others through AI stands outside the law made for equals and should be exiled. Exile, however, works only while the many can still remove the one, a power they would lack against such a person.

Later arguments for limiting government also rest on the presupposition. They answer the claim that whoever truly knows what is good should rule without restraint (Plato, c. 375 BCE) by pointing out that no actual ruler has been wise or capable enough to deserve unrestrained power. Rulers must therefore be removable without bloodshed, power must be divided so that its parts restrain one another, and the people must at least be able to judge who is fit to govern (Mill, 1859; Popper, 1945; Madison, 1788; Niebuhr, 1944; Montesquieu, 1748). The effort to build AI runs toward a person or a small group whose capability genuinely exceeds everyone else's across most domains, a case in which the answer loses its ground.

Divided power and democracy also depend on the second half of the presupposition, that the others can understand whoever holds power: each branch must be able to follow what the others are doing, and the people must be able to evaluate their governors. If governing came to rest on AI-assisted decisions that neither rival officials nor the public could evaluate, divided power and democracy would lose their footing. So would defenses of democracy that rely on the combined judgment of many citizens (Landemore, 2012; Estlund, 1997; Anderson, 2006).

Science

Science rests on the second half of the presupposition: its methods and institutions were designed by and for knowers with limits. Modern empirical science has been traced to the Christian belief that the human mind, damaged by the first sin, was too impaired to reach truth by reasoning alone, so that knowledge had to be built slowly, tested against experience, and corrected over time (Harrison, 2007). In the same spirit, hypotheses are proposed, tested, and replaced, so that no result is ever final (Popper, 1934; Kuhn, 1962; Lakatos, 1978).

In scientific institutions, work is divided among many people, results are reviewed by peers, credit goes to whoever was first, and trust in a result depends on others being able to follow it, because individual knowers have limited time, limited competence outside a specialty, and an ever-present possibility of error. None of the features assumes that one producer could work at a scale far beyond the community that receives the work.

Two questions should be kept apart: whether results produced by AI are true, and what would happen to scientific institutions if results arrived faster than any community

of human peers could follow them. If many results were wrong, the burden of finding the errors would fall on people who cannot work at the producer's speed. If the results were right, credit, priority, and the training of the next generation would be organized around work that human beings did not do. In both cases the institutions would keep operating, while the problem they were built to handle, the limits of each human knower, would give way to one they were not built for, a producer that no community of peers can follow.

If discovery capability were concentrated, it would widen the gap between those who command AI and those who do not. If it were widely diffused and applied to technologies that can serve both medicine and weapons, a few people could cause harm that no one could stop (Bostrom, 2019).

The Economy

Economics offers one of the clearest statements of the presupposition. The knowledge relevant to economic decisions is spread across many people and cannot be gathered in one place, so decisions are best left to many people coordinating through prices. The problem would reduce to calculation only for someone who held all the relevant information (Hayek, 1945). AI is being built to push back the limit of any single mind, toward the case in which the argument for leaving decisions to many people would lose its ground.

Economists have long explained value by scarcity relative to need (Menger, 1871). As AI reproduces cognitive products at ever lower cost, their value falls, while the land, energy, and computing capacity that AI runs on remain scarce. The fall has already begun, because when markets judge outputs and not the processes behind them, AI output that resembles skilled human work drives human producers out (Cao, 2026). Automation displaces workers from existing tasks while new tasks reinstate them, so long-run wages depend on whether new tasks keep appearing (Acemoglu and Restrepo, 2019; Korinek and Suh, 2024). A general system that matches or exceeds human beings across most domains, which is what AI is being built to become, would make such tasks rare. People who live by their work would then lose what others need from them, the second condition of comparability.

Trade offers some comfort, since even a less productive party gains from trade by specializing in what it does relatively best, a principle economists call *comparative advantage*. Because AI capacity is itself limited by computing power and energy, some work would remain for people. The principle guarantees gains from trade, but it does not guarantee that the gains cover the cost of living. In the long run it may stop applying altogether, once AI becomes so broadly effective and so cheap that an economy organized around human work no longer makes sense (Amodei, 2024).

Redistribution could in principle compensate those who lose from AI (Korinek and Stiglitz, 2017), provided that someone has both the means and a reason to tax the winners and pay the losers. Those who hold AI would have the means, but they would have a reason only as long as the many remained comparable to them, the condition that AI is removing.

Morality

Morality rests on the same presupposition. Most of what law and morality demand is restraint, above all the prohibition on killing and violence. Restraint is necessary because human beings can be hurt by one another, and it is possible because none of them can dominate the rest for long (Hart, 1994; Hobbes, 1651). Peace among human beings rests on their acknowledging one another as equals, whether or not nature made them equal (Hobbes, 1651), so moral restraint depends on the stronger party regarding the weaker as standing on the same level. When the stronger party comes to regard the weaker as beings of a different kind, the restraint of justice lapses and only compassion is left (Hume, 1998; Hamilton, 1779). A real superiority of strength can be enough to produce such a belief (Hume, 1998).

Even the strongest human being must sleep and can then be overcome by others, so restraint pays even for the strong, while a person far stronger than everyone else who never needed rest would gain more by attack than by restraint (Hart, 1994). The effort to build AI runs toward such a position: a real superiority of strength, held by those who command AI over those who do not, in a form that needs no rest. If those who command AI came to regard those who do not as beings of a different kind, the restraint of justice would lapse in the same way.

The Third-Category Entity

In the Christian tradition that shaped Western thought, the presupposition took a religious form: an infinite God stood apart from finite human beings, and among those acting in the world, everyone was finite. God is what reinforcement learning calls the environment, the rules that fix what can be done and what follows from each action, and does not act as one cause among others within it (Aquinas, 1265–1274). A covenant, a binding promise between God and human beings, linked the two positions and gave finite lives a calling and a measure for approaching the good.

Everyone who tried to occupy a position close to God's, from pharaohs to kings, revealed their finitude in the end. Their authority needed bloodlines, offices, or institutions to outlast them, and even authority resting on extraordinary personal qualities, which sociologists call *charisma*, could not survive the leader's death without being turned into ordinary rule (Weber, 1922).

AI differs from every historical superhuman authority in three respects. Its capability is real and does not rest on perception or faith. It does not die and so faces no problem of succession. It is not human and cannot be absorbed into institutions designed for humans. AI acts within the world as one cause among others, so it cannot occupy the position of God. The three differences keep it out of the position of finite human beings, which makes it what we call a *third-category entity*.

AI is being built to do what human beings do and to do it better, so that human finitude would become a plain disadvantage (Anders, 2025). God's infinity gave finite lives their meaning through the covenant, while the infinity AI is being built toward would leave human beings without meaning, because AI stands in no relationship with them that confers meaning and would only replace them.

No Landing Point

Whether a new order can be built after the presupposition fails can be judged against the last crisis of similar depth in European history, the religious wars that followed the split of Western Christianity, which ended only when exhausted rivals of roughly equal strength had to settle (MacCulloch, 2004). Rough equality arose there without design, because the parties were made of the same human material and shared the same limits. A gap of the kind considered here would leave no domain in which such equality still held and on which a settlement could rest.

Past orders stayed stable, and liberty survived, only while the parties remained comparable (Acemoglu and Robinson, 2006, 2019). A stable system of tiered access to AI would leave the gap in place, because such a system would rest on the forbearance of the upper tier, which would not permit lower-tier access that truly threatened it. Using AI to understand and restrain those who hold it meets the same limit, because the AI available for the purpose is the AI the holders release.

Proposals made within existing institutions assume that people can understand and evaluate what AI does. Current AI rules build on the assumption, asking for human oversight of AI (National Institute of Standards and Technology, 2023) and, in the strictest case, requiring that the overseers be able to understand the system, interpret its output, and override it (EU AI Act). Once AI's judgment exceeded that of the people it serves, such oversight, including oversight of AI that acts on people's behalf, would resemble a clerk signing documents the clerk cannot read.

Concentration and diffusion, the two forms the gap can take, lead to different ends (Bullock et al., 2025). Neither end can be undone by reform, because reform requires parties that remain comparable. Under concentration, the result would be one or more centers of unchecked power, a path toward dictatorship. Unlike earlier ruling minorities, which depended on the many beneath them (Mosca, 1939), holders who no longer needed the many would exclude everyone else even if several of them balanced one another.

Under diffusion, in domains where a technology is both very destructive and easy to use, a small number of people willing to cause harm is enough to make devastation likely, beyond the reach of deterrence or of protective technology developed in advance (Bostrom, 2019). We take as a premise that such domains exist. Because AI is general, released capability cannot be confined to domains where protection keeps the advantage, so wide diffusion of AI would lead toward lasting instability. The remedies proposed for such a world depend on institutions that do not yet exist (Bostrom, 2024).

Competition among the holders does not force the upper tier to open access, because rivalry among rulers worked in their subjects' favor only while each rival needed its subjects to fight, pay taxes, and work (Tilly, 1990), needs that AI is coming to supply (Drago and Laine, 2025). Competition decides only which of the two ends is reached: the further the frontier pulls ahead of what is released, the closer the world comes to concentration, and the more rivals release in order to resist a holder who is pulling ahead, the closer it comes to diffusion.

Because reform requires comparable parties, competition decides only which end is reached, and no domain keeps the equality on which a settlement could rest, institutions built on rough equality among people leave no outcome other than lasting instability or

dictatorship.

We have assumed that those who hold AI can direct it. If the holders themselves cannot evaluate what their systems do, no human being is fully in control, and the picture approaches the one drawn in work on gradual disempowerment (Kulveit et al., 2025), which the present analysis then meets from a different starting point.

Conclusion

What Amodei hopes AI will bring, the defeat of most infectious diseases and rapid economic growth (Amodei, 2024), depends on voters being able to evaluate how AI is used, on human peers being able to verify results, and on markets and taxation continuing to spread the benefits. All three rest on the presupposition, so the hope depends on what the effort to build AI is running toward removing.

Avoiding both dictatorship and lasting instability would require institutions that do not presuppose rough equality among people, which human civilization has not yet built and which may not be possible. Recognizing that the problem lies at the level of the presupposition shared across domains is a precondition for looking for them.

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